

Standard Operating Procedure

Tax & Cost Basis Data Quality Review

Version: 2.0

Last Updated: 09/27/2026

Owner: Tax & Cost Basis Operations Team

Purpose

This SOP defines the standard process for conducting monthly data quality reviews of securities transaction data to ensure accurate cost basis calculation and IRS tax reporting compliance.

Scope

Applies to all transaction data processed through the Tax & Cost Basis platform including BUY/SELL transactions, cost basis lot assignments, and 1099-B tax reporting records.

Pipeline Overview

This process runs in two phases:

Phase 1 — Data Setup (Run once per data refresh)

```
bash setup_data.sh
```

Generates synthetic transaction data and loads all three tables into PostgreSQL (apex_tax_db).

Phase 2 — Analysis Pipeline (Run monthly or on demand)

```
bash run_pipeline.sh
```

Executes all SQL checks, advanced analysis, visualizations, and AI-generated management reports in sequence.

Data Model

Three tables support the full review process:

- **transactions** — BUY/SELL events across tracked securities
 - **cost_basis_lots** — acquisition cost per lot (FIFO / LIFO / Specific Identification)
 - **tax_reporting** — realized gain/loss output feeding 1099-B reporting
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Process Steps

Step 1 — Run Data Quality Checks (Day 1-2 of month)

- Execute `sql/03_data_quality_checks.sql` via PSQL or `run_pipeline.sh`
- Results auto-saved to `reports/quality_check_results.txt`
- Six checks covering: null detection, calculation validation, status exceptions, discrepancy analysis, gain/loss validation, monthly trend analysis
- Flag any variance > \$0.01 between calculated and recorded amounts
- **Note:** PostgreSQL requires `::numeric` casting on financial calculations — applied throughout all SQL files

Step 2 — Run Advanced Analysis (Day 2)

- Execute `sql/04_advanced_analysis.sql` via PSQL or `run_pipeline.sh`
- Results auto-saved to `reports/advanced_analysis_results.txt`
- Covers: running totals, price deviation outlier detection, FIFO cost basis simulation, account-level exception scoring, month-over-month trend analysis

Step 3 — Triage Issues By Priority

Priority 1 — Immediate Action Required:

- Missing cost basis on tax reports
- Gain/loss calculation variances
- ERROR status transactions above \$10,000
- Price outliers flagged by z-score deviation > 2.0

Priority 2 — Resolve Within 5 Business Days:

- PENDING transactions older than 3 days
- Discrepancies in tax reporting status
- Missing price data on transactions

Priority 3 — Monitor:

- Zero quantity transactions
- Unreported records approaching IRS deadline
- Accounts flagged as HIGH PRIORITY in exception scoring

Step 4 — Generate Management Report (Day 3)

- `run_pipeline.sh` executes notebooks automatically
- `notebooks/03_analysis.ipynb` — produces Plotly visualizations saved to `visualizations/`
- `notebooks/04_ai_analysis.ipynb` — generates executive summary via Claude AI API saved to `reports/management_report.md`
- `notebooks/05_ai_deep_analysis.ipynb` — generates deep-dive analysis on highest-risk areas saved to `reports/management_report_detailed.md`
- Distribute reports to Operations Manager by end of Day 3

Step 5 — Resolution Tracking

- Log all issues in the Exception Tracking Report
- Update status daily until resolved
- Escalate unresolved Priority 1 items to team lead after 24 hours
- Cross-reference account-level exception scores to prioritize remediation order

Step 6 — Month End Sign-Off

- Confirm all Priority 1 issues resolved
- Validate IRS reporting status for all SELL transactions
- Verify holding period classifications (SHORT vs LONG term — tax rate implications)
- Archive monthly reports and issue log to `reports/` folder

Key Technical Notes

- All SQL files use explicit `::numeric` casting for financial calculations
- Date columns require `::DATE` casting when loaded via pandas to avoid text/DATE type mismatches
- Pipeline runs from project root — all outputs route to correct subfolders automatically
- AI reports use multi-turn Claude API conversations for deeper analysis beyond summary reporting
- Shell scripts use `set -e` — pipeline stops automatically if any step fails

Key Contacts

- Operations Manager: [Name]
- Tax Reporting Lead: [Name]
- Product Team: [Name]

Related Documents

- `sql/03_data_quality_checks.sql`
- `sql/04_advanced_analysis.sql`
- `setup_data.sh` — data setup pipeline
- `run_pipeline.sh` — analysis pipeline
- IRS 1099-B Filing Guidelines